

Final Project

Next-Gen NLP:

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Introduction

Artificial Intelligence and Industry Transformation

Artificial Intelligence and Industry Transformation

- Artificial Intelligence is transforming industries by enabling **automation, improved decision-making, and new technologies**.
- Companies across sectors are investing in AI to **increase productivity, reduce costs, and gain competitive advantages**.
- However, the **impact of AI varies across industries and companies**.
- News media provides insights into **how AI developments are discussed, adopted, and perceived**.

Project Context

- Analysis of **~200,000 news articles** related to **AI, machine learning, and data science**.

Research Questions

- Which industries and companies are most likely to be impacted by AI over the next several years?
- How are these industries and companies impacted by AI? (positive, negative, or uncertain impact)
- What factors appear to influence successful or unsuccessful AI adoption?

Data

Dataset Overview

Source: AI related news content

Dataset size:

- ~200,000 news articles

Key columns used in analysis:

Column	Description
Text	Full article content
Date	Publication date
Language	Article Language

Why this dataset?

- Large corpus of real-world news coverage
- Captures how AI developments are discussed across industries and companies

Article Cleaning & Filtering

1. Duplicate Removal

- Removed duplicate articles based on the `text` column. Prevents repeated articles from biasing the analysis

2. Article Length Filtering

- Created column:
`text_length = length of article text`
- Filtered articles to remove:
 - very short articles (low information)
 - extremely long articles (possible noise)

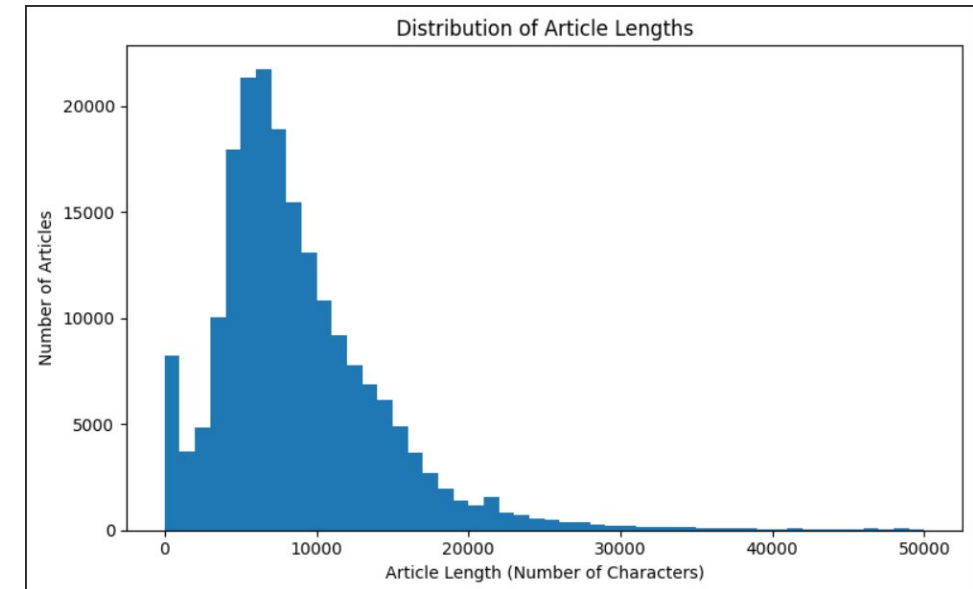
3. Relevance Filtering

Based on presence of:

- Created a relevance score:
 - AI-related keywords
 - impact-related keywords

`rel_score`

Filtering rule : `rel_score ≥ 3`. This keeps articles discussing **AI impact on industries and companies**.



Text Preprocessing

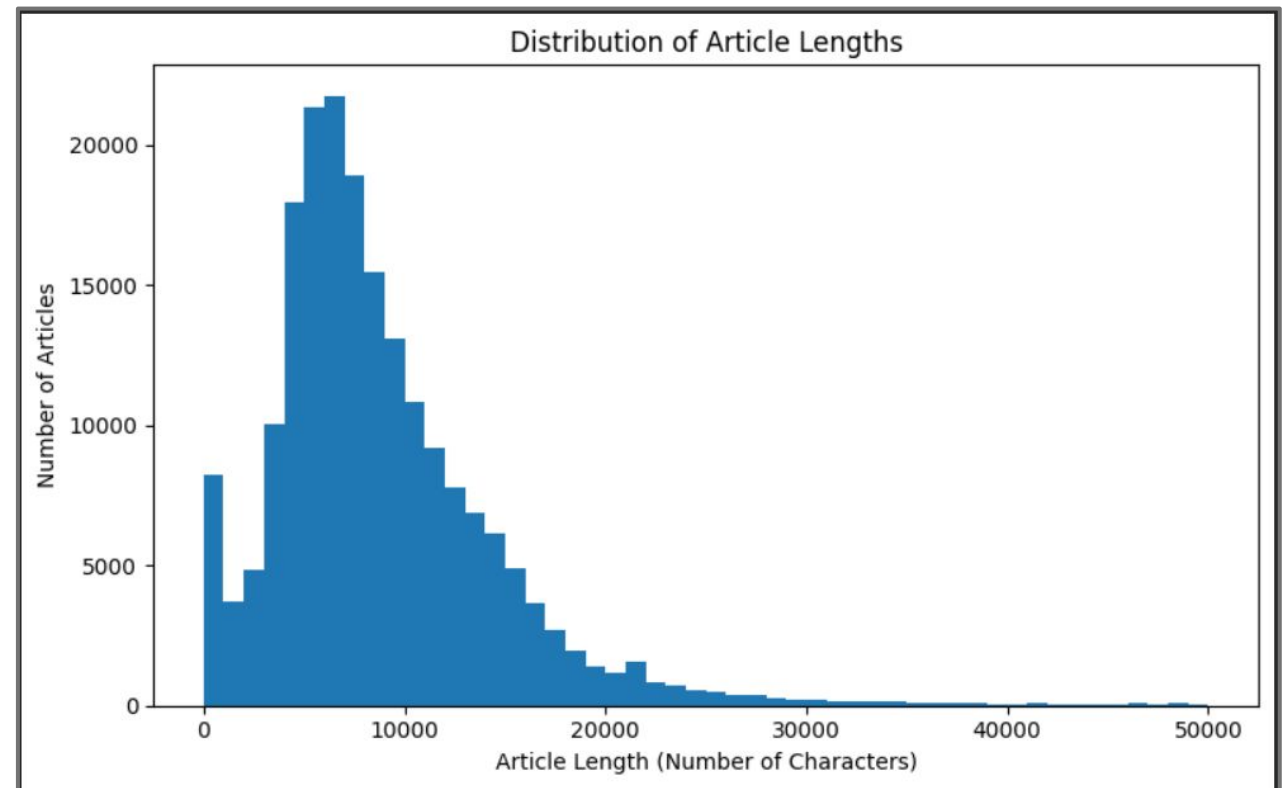
Before applying NLP models, the article text was standardized. Create column: `clean_text`

Processing steps:

- Convert text to lowercase
- Remove punctuation and numbers
- Remove extra whitespace

Purpose:

- Standardize text across articles
Reduces noise in the text data.
- Prepare text for feature extraction and topic modeling.



Feature Engineering (TF-IDF)

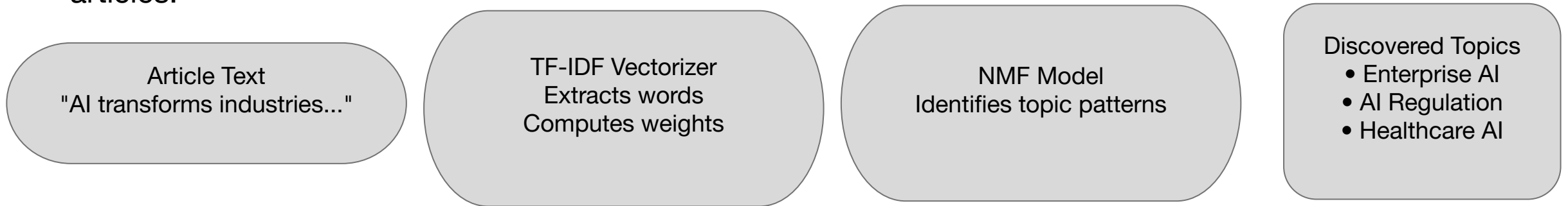
- To represent text numerically, I applied **TF-IDF vectorization** to the cleaned articles.
- **Input:**
clean_text
- **Output:**
TF-IDF document-term matrix
- **What TF-IDF measures**
- **Term Frequency (TF)**
 - How often a word appears in a document
- **Inverse Document Frequency (IDF)**
 - Reduces weight of words that appear in many documents

Why TF-IDF?

- Highlights **informative words** in articles
- Reduces influence of very common words
- Creates numerical features for **topic modeling**

Topic Detection (NMF)

- Applied Non-negative Matrix Factorization (NMF) to TF-IDF features to identify major themes in the articles.



→ **Model Input**

- TF-IDF document-term matrix.

Model Output

- **topic_id** — dominant topic assigned to each article
- **topic_weight** — strength of that topic in the article.

- Each topic represents a **cluster of articles discussing similar themes related to AI.**

? Why NMF ?

- Works well with **TF-IDF features**
- Produces **interpretable topics**
- Groups articles based on **similar word patterns**

Topic Interpretation

Top words from each NMF topic were examined to assign interpretable topic labels.

Topic	Label
Topic 0	AI Policy, Regulation & Public Opinion
Topic 1	AI in Healthcare & Compliance
Topic 2	Enterprise AI Platforms
Topic 3	AI Stocks & Market Investing
Topic 4	AI Startups & Product Launches (Press Releases)
Topic 5	Global AI Business & Impact Narratives (Asia/India, Bubble Fears, Risk)
Topic 6	Big Tech GenAI Products (Apple/Google/Gemini/ChatGPT)
Topic 7	Enterprise Adoption & Digital Transformation (Deployments/Innovation)
Topic 8	Noise (Currency Conversion / Spam) – DROP
Topic 9	OpenAI/ Microsoft Leadership & Company News

Topic labels were assigned by examining the top weighted keywords from each NMF topic.

Entity Extraction

To identify companies and technologies mentioned in the articles, I applied **Named Entity Recognition (NER)** using the **spaCy NLP library**.

Entities extracted:

- **org_entities** — organizations and companies
- **product_entities** — technologies and AI products

Purpose

Entity extraction allows us to:

- Identify **which companies are most frequently associated with AI developments**
- Link companies to **specific AI topics and sentiment patterns**

Example Entities Detected

Examples of organizations frequently mentioned:

- Microsoft
 - Google
 - Nvidia
 - Amazon
 - OpenAI
- These companies appear frequently in discussions of **AI development and adoption**.

Most Mentioned Companies

Org_entities	Count	Org_entities	Count
Microsoft	10172	npr	2065
Google	8507	the u s	2011
IBM	4734	fcc	1934
gray television inc	3326	un	1823
gpt	3318	gray media group inc	1797
fda	3017	microsoft s	1713
intel	2745	pbs	1623
congress	2705	cnn	1539
sec	2475	android	1476
samsung	2360	cbs	1458

Supervised Sentiment Model

Training Data

- A labeled sentiment dataset was used to train the Logistic Regression classifier. (positive / negative text examples)
- Dataset used to train a sentiment classifier

Model

- Logistic Regression classifier trained on TF-IDF features

Pipeline

1. Clean text
2. TF-IDF vectorization
3. Logistic Regression sentiment classifier
4. Predict sentiment for AI news articles

Output

- Predicted sentiment label
- Positive / Neutral / Negative

Model evaluation on validation set:

Accuracy: 0.87

F1-score: 0.85

Model Performance

- Logistic Regression classifier achieved **~87% accuracy on labeled sentiment data**, indicating reliable sentiment classification performance.

Sentiment Analysis Method

Method: Sentiment classification was performed using a **trained Logistic Regression classifier on TF-IDF features**.

Output: Two variables were created:

sentiment_score

Predicted sentiment score produced by classifier

sentiment_label

Final sentiment category assigned to each article

(Positive, Neutral, Negative)

Purpose

Sentiment analysis helps evaluate:

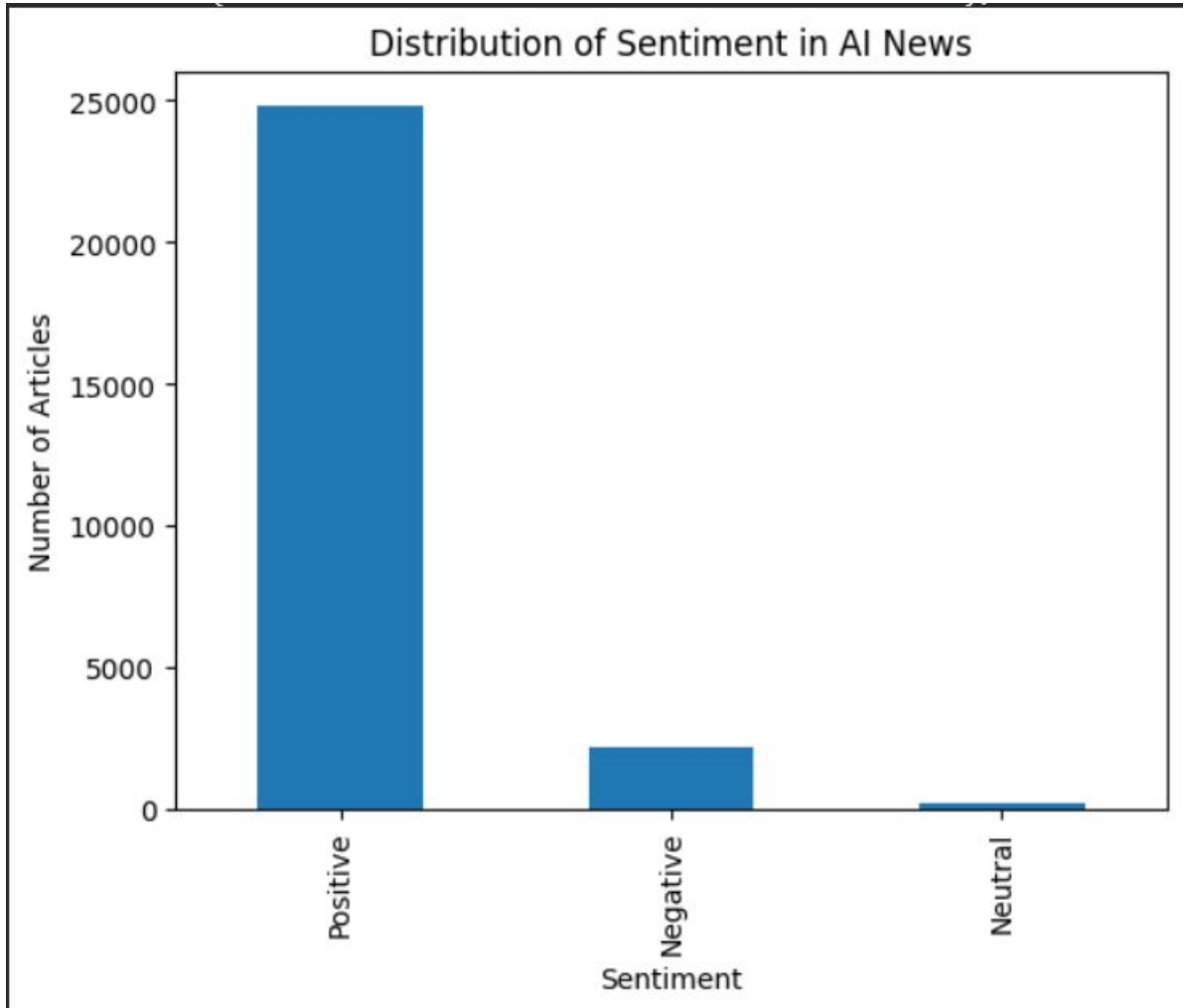
- whether **AI developments are portrayed positively or negatively** in the media
- how sentiment varies across **industries, companies, and topics**

Example:

"AI technology is transforming healthcare"
→ predicted sentiment = Positive

Predicted Sentiment	Description
Positive	Article Expresses Optimistic or favourable view of AI
Neutral	Article presents balanced or factual discussion
Negative	Article highlights risks, regulation, or concerns

Overall Sentiment Distribution



Method: Sentiment labels were predicted using the trained Logistic Regression classifier applied to the TF-IDF text features.

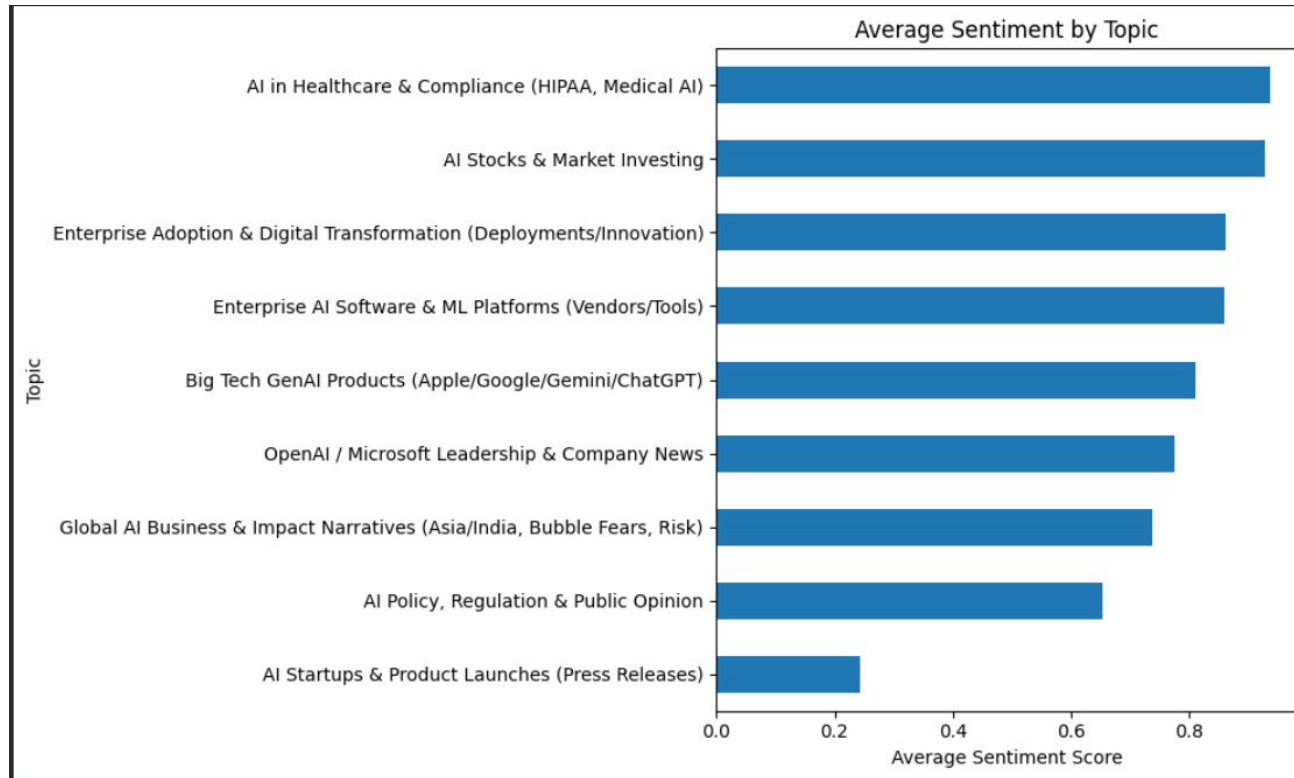
Articles were classified as:

- **Positive**
- **Neutral**
- **Negative**
This allows us to measure **how AI developments are portrayed in news coverage.**

Key Findings:

- Most AI-related news articles are **neutral or positive.**
- Positive sentiment often appears in discussions about:
 - **AI innovation**
 - **technological advancements**
 - **business adoption**

Topic-Level Sentiment



Method

To understand how AI is discussed across different themes, sentiment scores were **aggregated by topic**.

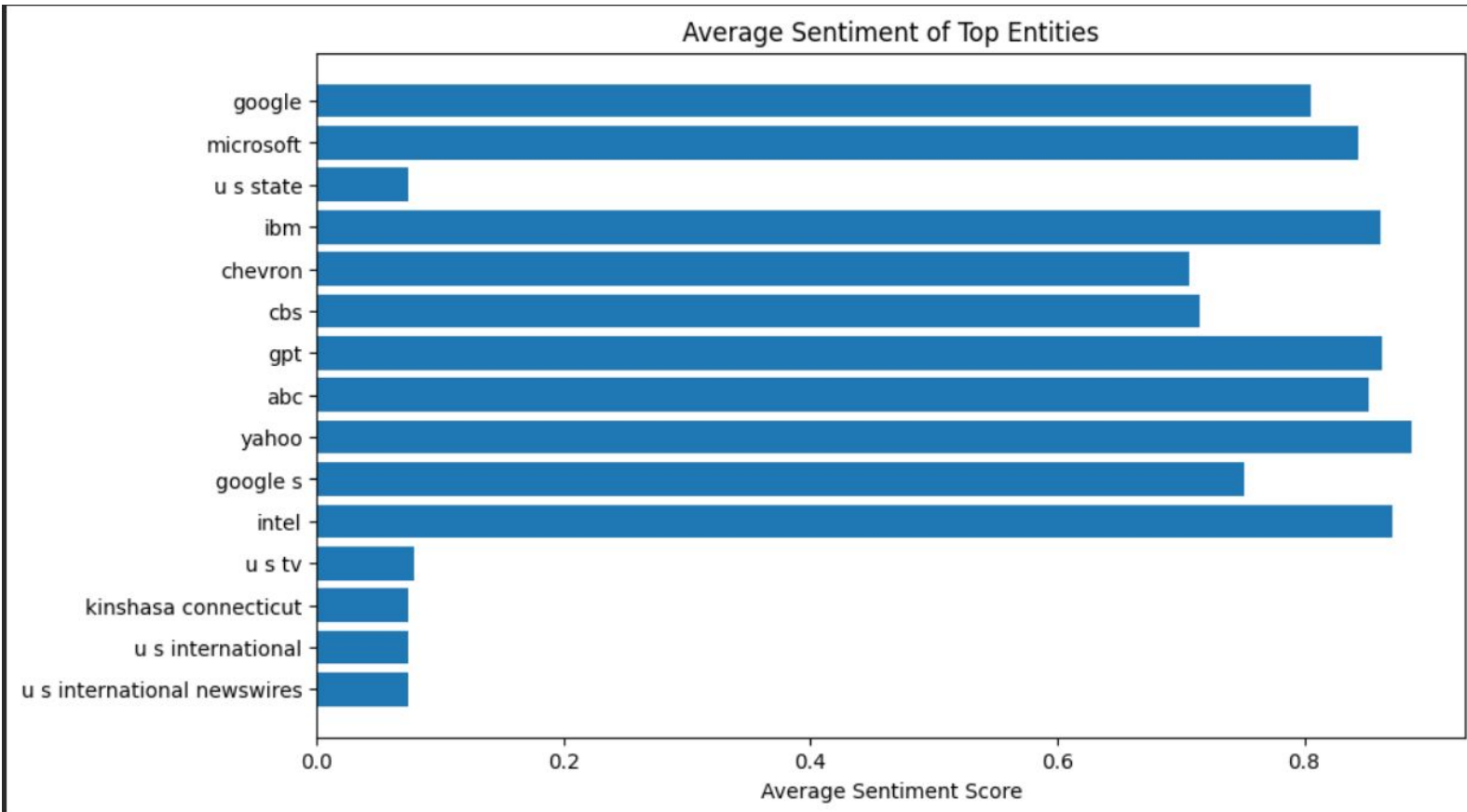
Metrics analyzed:

- Average sentiment score
- Percentage of positive articles
- Percentage of negative articles

Key Observations

- Topics related to **AI innovation and product development** show **more positive sentiment**.
- Topics related to **regulation, risks, and workforce impact** tend to show **more negative sentiment**.
- Many topics remain **neutral**, reflecting balanced news coverage of AI.

Entity-Level Sentiment



→ Method

Sentiment scores were linked with **organizations** extracted using **Named Entity Recognition (NER)**.

This allows us to evaluate how companies involved in AI development are portrayed in news coverage

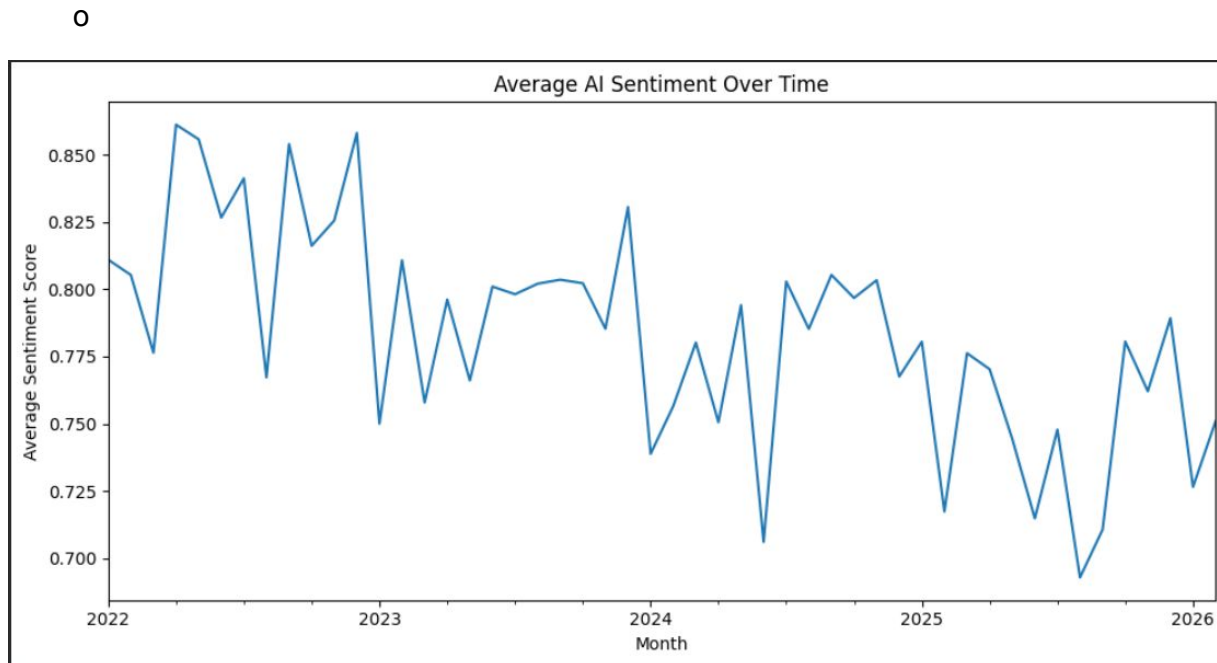
→ Key Findings

- Major technology companies such as **Microsoft, Google, and IBM** tend to appear in articles with **more positive sentiment**.
- Discussions involving **government institutions or regulatory bodies** show **more neutral sentiment**.
- AI innovation and product development are generally portrayed **positively in news coverage**.

→ Interpretation

Companies leading AI innovation tend to receive **more positive media coverage**, while discussions involving **regulation, competition, or ethical risks** produce more neutral or negative sentiment.

Sentiment Over Time



Method

Sentiment scores were aggregated **by month using the article publication date** to track how AI sentiment changes over time.

Key Findings

- Overall sentiment toward AI remains **mostly neutral to positive**.
- Negative sentiment spikes **often correspond to discussions about:**
 - regulation
 - ethical concerns
 - workforce disruption
- **Positive sentiment increases** during periods of:
 - AI innovation
 - major product announcements

Key Findings

1. AI discussions are dominated by large technology companies

- Companies such as **Microsoft, Google, Nvidia, and Amazon** appear most frequently in AI-related news articles.

2. AI innovation topics tend to have more positive sentiment

- Articles discussing **AI development, products, and innovation** often show positive sentiment.

3. Regulation and workforce concerns show more negative sentiment

- Topics involving **policy, ethics, and job displacement** tend to have more negative or cautious sentiment.

4. AI adoption varies across industries

- Some industries focus on **product innovation**, while others emphasize **regulation and societal impact**.

AI Adoption Insights

The analysis of topics, entities, and sentiment suggests several factors that influence **AI adoption across industries**.

Drivers of Successful AI Adoption

- Clear **business use cases** for AI technologies
- Strong **data infrastructure and technical capabilities**
- Integration of AI into **existing workflows and decision-making**

Challenges to AI Adoption

- **Regulatory and policy uncertainty**
- Concerns about **ethical implications and job displacement**
- High **implementation complexity and cost**

Interpretation

News coverage suggests that organizations with strong technological capabilities and clear AI strategies are more likely to **successfully adopt AI technologies**.

Conclusion & Q&A

This project analyzed **~200,000 news articles** to understand how AI is discussed across industries.

Key takeaways:

- AI discussions are dominated by **major technology companies**
- Innovation-focused topics show **more positive sentiment**
- Regulatory and societal concerns introduce **more cautious or negative sentiment**

Overall, AI is transforming industries, but **its impact varies depending on context and industry.**

This analysis shows that AI discussions are dominated by major technology firms, innovation topics tend to receive positive sentiment, and adoption success appears linked to strong infrastructure, clear use cases, and regulatory context.